



Association of sequence types, antimicrobial resistance and virulence genes in Indian isolates of *Klebsiella pneumoniae*: A comparative genomics study

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ABSTRACT

Objectives: *Klebsiella pneumoniae* genomic surveillance in India was performed to understand the spread of antimicrobial resistant genes across the country over 10 years and to delineate environment and clinical *K. pneumoniae* through comparative genomics approach.

Methods: The genomic data of 153 strains were retrieved from PATRIC. To compare the presence of AMR genes, virulence genes, episomes and their evolutionary relationship in 153 *K. pneumoniae* genomes, Res-Finder, Virulence Factor Database, Plasmid Finder and CSI phylogeny tools were used, respectively.

Results: Diverse serotypes were found among the 153 *K. pneumoniae* isolates, of which K51 (28%) and K64 (21.56%) were majorly found. Most of the K51 isolates belong to ST231 (93.02 %). The number of associated virulence genes (*rpmA*, *magA*, *entB*, *ybtS*, *iutA*, *allS*) appeared to be higher in ST231-KL51 and ST23-KL1 isolates. More than 97% of clinical strains have yersiniabactin, aerobactin genes. Importantly, 98% and 62% of the ESBL and carbapenemase-producing isolates harboured *ybtS*, *iutA*, and *rpmA*, *magA* genes, respectively. The IncF conjugative plasmids are predominant in *K. pneumoniae*. The phylogenetic analysis clearly separates the environmental strains from clinical and characterised by uncommon STs and serotypes.

Conclusion: Our study illustrates *K. pneumoniae* genomic surveillance in India representing the phylogenetic evolution, STs, AMR, virulence, serotype to provide more attention for immediate treatment and preventing the dissemination of *K. pneumoniae*.

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1. Introduction

Klebsiella pneumoniae has a strong association with human as resident flora and colonises almost every part of the human body with most preferential in the respiratory, gastrointestinal and urinary tract [1]. The other predominant habitats include soil, plants,

surface water, sewage and industrial effluent [2]. Despite the source of origin, clinical isolates cause nosocomial and community-acquired infections. This leads to the serious therapeutic threats given the increase in drug resistant phenotypes and decrease in effective antibiotics [3]. The emergence and spread of extensive drug-resistant *K. pneumoniae* are a global concern, which causes severe untreatable infections in immunocompromised patients [4]. The associated character of antibiotic resistance and virulence in *K. pneumoniae* was reported to have high morbidity and mortality rates in immunocompetent individuals as well [5]. Pneumonia, meningitis, urinary tract infections and bloodstream infections are the potential illness caused by this bacterium in immunocompromised [1]. To treat these infections, similar classes of antibiotics have been in use for the last two decades that render the bac-

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terium to evolve strategies to survive in an antibiotic stress environment [6]. Few potential survival strategies include acquisition of new antimicrobial resistance genes (AMR), production of hydrolytic enzymes, expression of efflux proteins and formation of biofilm [7,8]. The majority of AMR genes are acquired through conjugation, transformation and transduction; hence understanding the co-occurrence of plasmid type and AMR gene transfer could help to limit the dissemination of AMR and virulence genes. Nevertheless, antibiotic resistance is not confined to clinical strains alone; it is also widespread among the non-pathogenic environmental isolates [9]. Some important virulence factors of *K. pneumoniae* include capsular polysaccharide, siderophores and fimbriae [10]. However, a few factors are common in both clinical and environmental isolates such as capsule, type 1 fimbriae, type 3 fimbriae and siderophores.

K. pneumoniae are classified into classical strain (cKp) and hypervirulent strain (hvKp) [11]. The hvKp is significantly different from the cKp by harbouring *rmpA* and *rmpA2* mucoid-regulator genes, K1, K2, K20 capsular types and aerobactin [11]. The hvKp strains can cause serious infections such as liver abscess and meningitis in both immunocompetent and immunocompromised individuals [12]. Also, hvKp strains are characterised by increased resistance to host defence mechanism and associated with high mortality rates, especially by the hypervirulent carbapenem-resistant *K. pneumoniae* (hvCRKP) [13]. This pathogen has emerged in the past decade, and the incidence of carbapenemase-acquired hypervirulent strain has been increasing in recent years. The convergence of carbapenem-resistant and hypervirulence poses a great threat as it is concomitantly multidrug-resistant, hypervirulent and transmissible [14,15]. Both cKp and hvKp are known to possess a high diversity of antimicrobial resistance genes, with increased resistance towards third-generation cephalosporin [16]. The combination of drug resistance and hypervirulence advocates *K. pneumoniae* as a clinically significant pathogen of global health concern. Also, *K. pneumoniae* is listed in the World Health Organisation (WHO) priority critical pathogens. As the hypervirulent and drug-resistant population remains to overlap in most isolates, it is necessary to identify the distinguishing features of pathotypes and environmental isolates of *K. pneumoniae* to prevent the spread of pathogenic strains in hospital and community settings.

The traditional taxonomic techniques 16S gene Sanger sequencing and PCR-based sequence typing (MLST) are all well-established techniques in characterising pathogenic bacteria [17–19]. However, these techniques are unable to distinguish closely related species as well as limited in detecting diverse AMR genes and virulome. Recent advancements in the next generation sequencing and extended applications of bioinformatic tools facilitate to gather information on thousands of bacterial species on their virulence genes, resistance genes and genetic relationship [20]. Comparative genomics of microbial genomes assists to understand the genomic variations, the basis of diverse phenotypes. Analysis such as antibiotic profiling, virulome analysis and plasmid types sheds light on the prevalence of drug-resistant genes, their mechanism, evolution and dissemination. With this ample significance, the comparative genomic investigations of *K. pneumoniae* of Indian origin were studied. We retrieved 153 genomes of clinical and environmental *K. pneumoniae* isolates of Indian origin from PATRIC, and comparative genomic analyses were made to understand the spread of *K. pneumoniae*-resistant strains across the country over 10 years, during which more whole genome of *K. pneumoniae* were sequenced. Then sequence types (ST), evolutionary relationships, the resistance mechanisms, acquisition of AMR genes, presence of conjugative and non-conjugative plasmids and virulence genes were screened. In addition, association among the four major factors such as virulence, resistance determinants, replicon types and ecological niche were studied for 153 genomes of *K. pneumoniae*. The genomic

surveillance is conducted to gain insights into genotypic characterisation which may facilitate immediate treatment and attention to prevent the dissemination in the environment and clinical settings.

2. Materials and methods

2.1. Retrieval of 153 genome sequence data

The whole genome sequence (WGS) of good-quality Indian isolates (n=153) of *K. pneumoniae* were collected from the PATRIC database available during our study period January to March 2021 (the Pathosystems Resource Integration Center) [21]. We retrieved the assembled FASTA sequence for 153 strains, which were previously assembled using Spades. These strains were sequenced using the Illumina, Nanopore and PacBio sequencing platform and shown in Supplementary Table S1. The sequence type, year of strain isolation and sampling source of the 153 isolates were collected from the PATRIC. Also, the genome size and GC content of each genome was retrieved. For further analysis, FASTA format was used.

2.2. Antimicrobial resistance profiling

AMR profile of each isolate were carried out using Resfinder 4.1 in Center for Genomic Epidemiology (CGE) database [22]. A total of eight antibiotic classes such as β -lactam, aminoglycoside, fluoroquinolone, fosfomycin, sulfonamide, macrolide, phenicol and rifamycin were selected. For AMR profiling, 88 AMR genes belonging to these classes were taken. Then the resistance mechanisms of 153 isolates were analysed using CARD (Comprehensive Antibiotic Resistance Database) [23].

2.3. Virulome analysis

Virulence Factor Database (VFDB) was used to screen the virulence genes. This tool allows the identification of type I fimbriae genes (*fimD*, *fimK*, *fimH*, *fimC*), type III fimbriae genes (*mrkD*, *mrkJ*, *mrkF*, *mrkC*, *mrkA*, *mrkI*), siderophore genes (*iutA*, *entB*, *ybtS*), allantoin utilization gene (*allS*), capsular genes (*wcaH*, *wcaG*, *manB*, *manC*, *hcaA*, *wcbR*, *rmpA* and *magA*), type IV pili (*tapQ*, *pilJ*), type III secretion system (*vopB2*) and type IV secretion system (*clpV/tssH*) [<http://www.mgc.ac.cn/VFs/>] [22].

2.4. Serotype analysis

To screen the type of serotype, Kaptive database was used [24]. Using this type K-locus types were predicted.

2.5. Plasmid profiling

Plasmid finder was used to identify plasmids that are found in genome sequence. The FASTA format was used to identify the replicon sequence that matched with 100% identity [25].

2.6. Phylogenetic analysis

The evolutionary relation among the 153 *K. pneumoniae* was determined using CSI phylogeny. In order to investigate the phylogenetic placement based on MLST, the seven housekeeping genes were concatenated for each genome and a reference strain, MGH 78578 [26] using MegaX. Neighbour joining method was used to view the tree with bootstrap value 1000. Tree editing and annotation were performed using interactive Tree of Life (iTOL) [27].

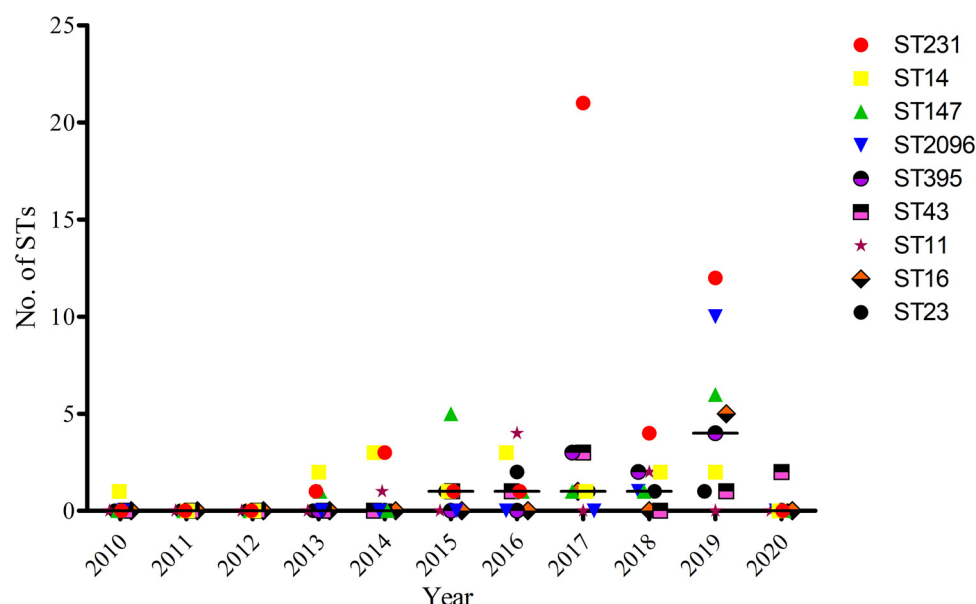


Fig. 1. Year-wise ST distribution: Scatter plot represents the distribution of STs across 2010–2020. ST231, ST147 and ST2096 are displayed with increasing trends during 2013, 2015 and 2018. The points represent the STs representing the abundance from high to less. The line denotes the median points.

2.7. Statistical analysis

The statistical tests were performed to delineate classical and hypervirulent strains based on AMR and virulence. The analysis was performed using RStudio version 1.41717, and data were visualised using ggplot2. Differences in total number of virulence and AMR genes among classical and hypervirulence categories were assessed using Wilcoxon test [28].

3. Results

3.1. Overview of the genomic metadata composition of *K. pneumoniae* isolates

The PATRIC metadata (Supplementary Table S1) depicts the isolation source, genome size, GC content, STs of 144 clinical strains and 9 environmental strains. The isolation source of 153 strains are blood (n=115), urine (n=9), endotracheal aspirate (n=5) bronchoalveolar lavage (n=4), sputum (n=4), pus (n=3), CSF (n=2), nasal swab (n=2), rhizosphere soil (n=1; AWD5), root nodule (n=1; HPCN22), domestic sewage (n=2; HPCN5, HPCN17), STP sludge (n=1; PVN-1), endophyte (n=1; ME30), river surface water (n=1; PL1-RCS238), industrial waste water (n=1; EGD-HP19) and agriculture field soil (n=1; KBG6.2). The genomic characterisation of *K. pneumoniae* was performed as first part of the study. From the PARTIC database, we found these genomes were sequenced majorly by Illumina and Ion torrent. The genomes of clinical strains (n=144) and environmental strains (n=9) ranged in size from 4.4 Mbp to 6.3 Mbp, and GC contents were between 55% and 57%.

3.2. Sequence type

The study isolates were found to have 37 different STs, suggesting their diversity in Indian isolates. We found ST231 was predominant (n=43) among clinical strains, followed by ST147 (n=15), ST14 (n=15), ST2096 (n=11), ST395 (n=9), ST43 (n=8), ST 11 (n=7), ST23 (n=6), ST16 (n=6), ST15 (n=3) and ST437 (n=2); other clinical strains comprised individual STs, ST13, ST42, ST101, ST307, ST557, ST570, ST660, ST628, ST711, ST2816, ST3249, ST3605, ST3607, ST3789, ST3835, ST3836, ST4847. The scatter plot represents the distribution of STs, where ST231, ST147 and ST2096

emerged and displayed with increasing trends during 2013, 2015 and 2018 (Fig. 1). Among these, ST14 was common across all years. The environmental strains composed of individual STs, namely, ST22, ST200, ST557, ST1107, ST1728, ST2701, ST3689. These STs, except ST1377, were also reported to be associated with clinical strains [29].

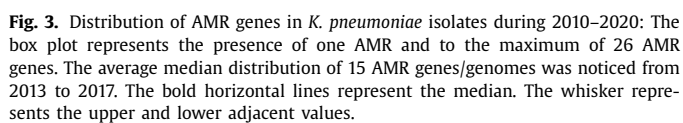
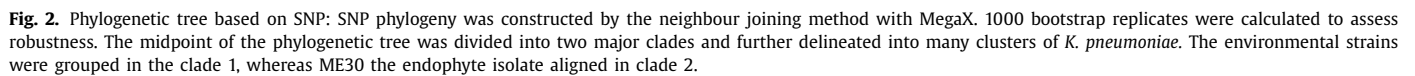
3.3. Phylogenetic analysis

The WGS of study strains were used for the phylogenetic analysis based on MLST (Supplementary Fig. S1) and single nucleotide polymorphism. The midpoint was divided into two major clades and further delineated into many clusters of *K. pneumoniae*. The environmental strains HPCN5, HPCN17, HPCN22, PL1-RCS238, AWD5, KBG6.2 and EGD-HP19 were grouped in the clade 1, whereas ME30 the endophyte isolate aligned in clade 2, suggesting the evolution as a separate ancestral group. The reference strain MGH78578 was closely aligned with environmental strain PVN1, STP sludge isolate. All the environmental strains were found in separate lineage from the clinical strains (Fig. 2).

3.4. *K. pneumoniae* has high prevalence of the genes conferring resistance to β -lactams, fluoroquinolones and aminoglycosides

Using the ResFinder, 88 different predominant AMR genes involved in eight antibiotic classes were selected for AMR profiling. The highest resistance genes found are fosfomycin 100% (n=153), β -lactam 96% (n=147) and fluoroquinolone 92% (n=142), followed by aminoglycoside 84% (n=129), sulfonamide 79.08% (n=121), phenicol 70 % (n=107), macrolide 64.04% (n=98) and rifamycin 48 % (n=74). AMR genes associated with fosfomycin (*fosA-like*), β -lactam (*bla_{CTX-M-15}*, *bla_{TEM-1B}*), fluoroquinolone (*oqxA-like*, *oqxB-like*) and aminoglycoside (*aadA2*, *aac(6')Ib-cr-like*) were predominant across the Indian strains (Supplementary Table S2).

To understand the distribution of AMR genes in 153 genomes during 2010–2020, box and whisker plot was used (Fig. 3). The data revealed the presence of the genome having at least one (n=1) and a maximum of 26 AMR genes (n=4). The average median distributions of 15 AMR genes/genome were noticed during 2013–2017.



The environmental strains also carried the *bla*_{SHV-1}, *fosA*-like, efflux pump mediated resistance (*oqx-A* like, *oqx-B*-like). Notably, the industrial effluent isolate carried *fosA*-like, *oqx-A*-like, *oqx-B*-like, *sul1*, *aac(6)lb-cr*-like, *bla*_{SHV112}, *ere (A)*-like, *aacA4*-like, *aadA1*.

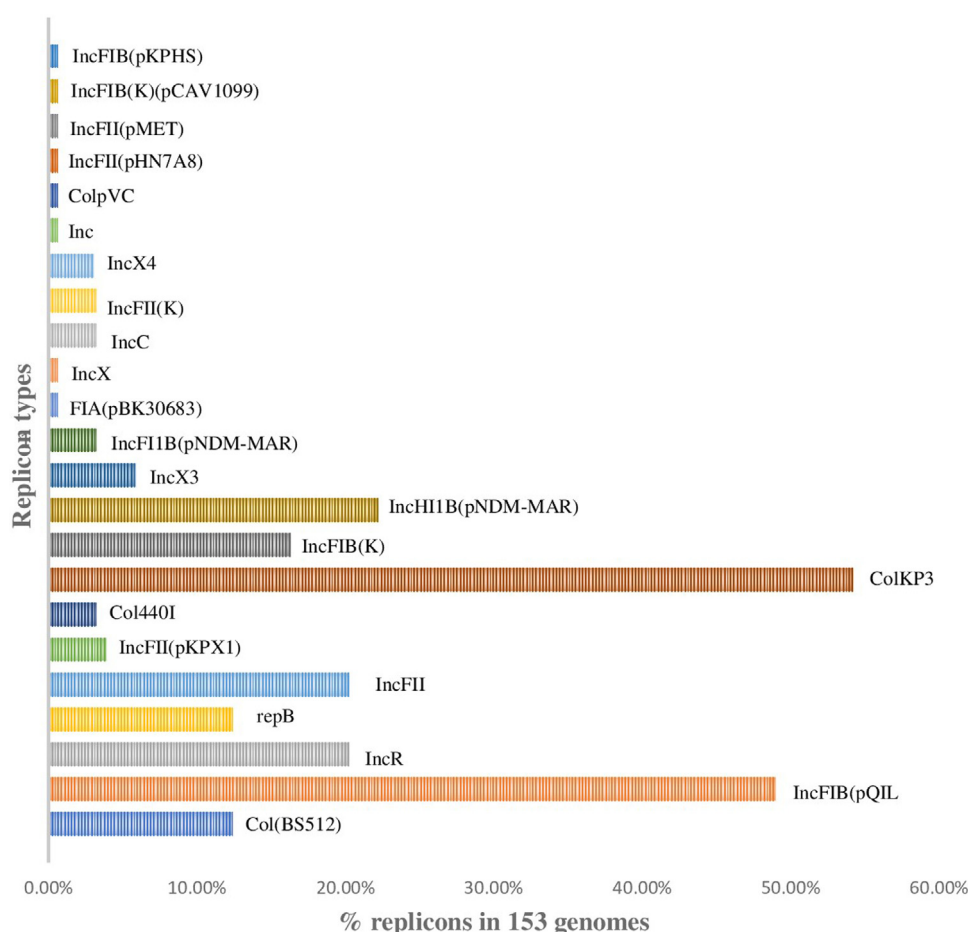


Fig. 4. Prevalence of replicon types among the study strains: The bar graph corresponds to the percentage distribution of replicon types among 153 genomes.

3.5. IncF-type conjugative plasmid is dominant among *Klebsiella*

To understand the transmission of AMR determinants through plasmids, the type of plasmid and the resistance genes they carry were investigated. From the metadata, the number of plasmids estimated to be ranged from one to eight plasmids per strain, with most isolates harbouring at least two plasmids. A clinical strain KpIMS38 isolated from pus harboured 16 plasmids. Then the episome details of 153 strains were obtained using Plasmid finder. A total of 19 plasmid replicon types were detected from 93.4% (n=143) of isolates. The 19 plasmid replicon types are categorised as incompatibility plasmids (Inc) and mobilizable colicin plasmid (Col) groups. The most abundant epidemic plasmid is IncF-type conjugative plasmid (80.39 % n=123) (Fig. 4). Plasmids belonging to IncF-type are reported to carry ESBL genes, genes encoding carbapenemases, aminoglycoside modifying enzymes and quinolone resistance genes [30]. The colKP3 plasmid was identified in 83 isolates, which is related to *bla*_{OXA-232} and *bla*_{OXA-181} transfer. The second most prevalent were pKpQIL-IT plasmids (n=75) belonging to IncFIB (pQIL) family, which carries an aminoglycoside resistance gene on a transposon-like element IS26. Both colKP3 and pKpQIL-IT plasmid were accompanied in 33.9% (n=52) isolates. These two dominant plasmids were not detected in any of the environmental strains. IncHI1B (pNDM-MAR) plasmid with NDM resistance gene was found in 33 clinical strains and an environmental strain. The colistin resistance genes (*mcr1*) encoding plasmids (IncX4, IncY) were detected only in three strains among 153 study isolates. Certain resistant genes *bla*_{CTX-M-15} (IncR, IncFII, IncFII, IncHI1B, Col440I) and *bla*_{NDM-5} (IncFII, IncX3) were detected

in various plasmid replicon types. In the case of environmental strains, five isolates were harboured with one plasmid. The endophyte isolate has two plasmids possessing *bla*_{CTX-M-15}, *bla*_{NDM}, *qnrB1*, *bla*_{SHV-56}, *fosA*, *oqxA* and *oqxB* resistant genes. The industrial effluent isolate has IncY, phage-like plasmids encoding *mcr1* gene; we have also recorded the multiple resistance being carried by this isolate. In summary, clinical and environmental strains have plasmids encoding genes that confer resistance to β -lactam drugs, aminoglycoside, quinolone and colistin. The type, family and functions of plasmids are discussed in Table 1.

3.6. Correlation of resistance and plasmid type

We narrowed down to identify the strains (n=15) that carry predominant AMR genes among the eight selected antibiotic classes and the presence of corresponding replicon types (Supplementary Table S4). These highly drug-resistant isolates were found to have pKpQIL-IT and pKP3-A plasmid type. pKpQIL-IT is found to harbour an aminoglycoside resistance gene on a transposon-like element IS26, which is a clinically important insertion sequence. pKP3-A, a mobilizable but non-self-conjugative plasmid, is associated with *bla*_{OXA-232} and *bla*_{OXA-181} genes, suggesting that pKpQIL-IT and pKP3-A replicon types has a significant role in the acquisition of AMR genes in *K. pneumoniae*.

3.7. Associated virulence genes are higher in ST231 strains

The presence of different virulence genes is shown in (Fig. 5), which were screened using VFDB. Both clinical and environmen-

Table 1
The replicon type, family and function.

Plasmid	Family	Antibiotic-resistant genes	Functions of the plasmid	References
pKpQIL-IT	IncFIB(pQil)	Aminoglycoside resistant gene	Plasmids capable of carrying transfer and MDR function	[51]
pK245	IncR	<i>bla</i> _{CTX-M-15}	Plasmid capable of carrying carbapenem resistance gene	[52]
pK2044	repB	<i>tetC</i> , <i>tetD</i>	Conjugal transfer and maintenance	[53]
pVCM01	IncFII	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{NDM-5} , <i>bla</i> _{TEM-1} , <i>bla</i> _{OXA-1}	Conjugal transfer and maintenance	[53]
pKPX-1	IncFII(pKPX1)	<i>bla</i> _{NDM-1} , <i>bla</i> _{CTX-M-15} , <i>bla</i> _{TEM-1} , <i>bla</i> _{OXA-1}	Conjugal transfer and maintenance	[54]
pKPN-IT	IncFIB(K)		Conjugal transfer and maintenance	[53]
pNDM-MAR	IncHI1B(pNDM-MAR)	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{NDM} , <i>qnrB1</i>	Limiting plasmid dissemination among potential hosts	[55]
pIncX-SHV	IncX3	<i>bla</i> _{NDM-5}	Carrying various carbapenemase resistance genes	[56]
pNDM-MAR	IncFII1B(pNDM-MAR)	<i>bla</i> _{NDM-1}	Limiting plasmid dissemination among potential	[55]
pT5282-CTXM	IncX5	<i>bla</i> _{KPC-5}	Dissemination	[57]
pNDM-KN	IncC	<i>bla</i> _{CMY} , <i>bla</i> _{NDM-1}	Broad host range plasmids capable of carrying transfer and MDR function	[58]
pKPN3	IncFII(K)	<i>bla</i> _{KPC-2}	Acquisition and dissemination of MDR	[59]
pCROD2	IncX4	<i>mcr-1</i>	MCR-1 carrying conjugative plasmid	[60]
P1 RepA (repA)	IncY	<i>mcr-1</i>	Phage-like plasmids	[61]
pHN7A8	IncFII(pHN7A8)	<i>bla</i> _{NDM-1}	Acquisition and dissemination of MDR	[62]
pMET-1 FC1	IncFII(pMET)	<i>bla</i> _{OXA-48}	Acquisition and dissemination of MDR	[63]
pCAV1099-114	IncFIB(K)(pCAV1099-114)	<i>bla</i> _{SHV-56} , <i>fosA</i> , <i>oqxA</i> , <i>oqxB</i>	Conjugal transfer and maintenance	[64]
pKPHS1	IncFIB(pKPHS1)	<i>bla</i> _{KPC} , <i>bla</i> _{OXA-48} , <i>bla</i> _{NDM-1}	Conjugal transfer and maintenance	[65]
pBK30683	FIA(pBK30683)	<i>bla</i> _{KPC-3}	Plasmids capable of carrying transfer and MDR function	[66]
FDAARGOS_440	Col440I	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{OXA-1} , <i>tet(B)</i> , <i>tet(D)</i> , <i>aac(6')-Ib-cr</i>	<i>bla</i> KPC bearing plasmids	[67]
pKP3-A	ColKP3	<i>bla</i> _{OXA-232} , <i>bla</i> _{OXA-181}	Carries carbapenems resistance genes (<i>oxa</i>)	[68,69]
pBS512_2	Col(BS512)	<i>bla</i> _{OXA-48} , <i>bla</i> _{NDM-1} , <i>bla</i> _{CTX-M-3}	Plasmid capable of carrying carbapenem resistance gene	[70]
pVCM01	ColpVC	<i>bla</i> _{OXA-181}	Hotspot bearing plasmid (fitness)	[71]

tal isolates were found to have genes involved in type 1 fimbriae (*fimD*, *fimK*, *fimH*, *fimC*). Similarly, most isolates were detected with *mrkD*, *mrkJ*, *mrkF*, *mrkC*, *mrkI*, *mrkA*, which are important genes for type 3 fimbriae. To identify the hypervirulent phenotypes, *rmpA* a gene regulator for mucoid phenotype was analysed. Nearly 47% (n=77) of clinical and 33.3% (n=3) of environmental strains were identified as hypervirulent strains. Furthermore, the prevalence of *magA* gene (mucoviscosity-associated gene) was comparatively less in clinical strains (n=56, 38.8%) than the *rmpA* genes. The *magA* was detected in environmental strains (n=5, 55.5%) as well. The presence of both *rmpA* and *magA* genes were not detected in root nodule, STP sludge and agriculture field soil isolate. Nevertheless, the co-occurrence of *rmpA* and *magA* gene was observed in domestic sewage isolates (n=2) and an industrial waste water isolate (n=1). Likewise, the *rmpA* and *magA* were accompanied in clinical strains (n=38). These clinical strains were identified as blood isolates majorly with K51 serotype (n=18). Among the study strains, K51 (n=43) and K64 (n=33) serotypes were majorly found. Yet K51 serotype was not associated with severe infections, and a recent report has identified K64 belonging to ST11 as an emerging superbug worldwide; however, none of the 153 Indian isolates belong to K64-ST11 [31].

The most reported hypervirulent K1 and K2 serotypes were found in a smaller number of strains, n=7 and n=11, respectively [32]. The co-occurrence of *rmpA* and *magA* genes were observed in K1 (n=6) and K2 (n=2) serotypes belonging to ST23 and ST14, respectively.

The three most important siderophores systems in Enterobacteriaceae are yersiniabactin (*ybtS*), aerobactin (*iutA*) and enterobactin (*entB*) were analysed in *K. pneumoniae* strains. The *entB* was detected in all the studied strains (n= 153); *ybtS* gene was found in 98.6% in clinical (n=142) and 77.7 % in environmental strains (n=7), and *iutA* gene was observed in 97.2 % in clinical and 88.8% of environmental strains. The root nodule strain was negative for both *ybtS* and *iutA*. In addition, allantoin utilisation (*allS*) was screened in the studied strains as it has been associated with hypervirulence [13]. The *allS* gene was observed in 65.97% of clinical (n=95) and 77.77 % of environmental strains (n=7). Further, the strains associated with hypervirulence factors (*magA*, *rmpA*, *allS*,

entB, *ybtS*, *iutA*) were identified in three environmental strains (EGDHP19, HPCN17 and HPCN5) and 29 clinical strains. The number of associated virulence genes appeared to be high in strains ST231- K51 and ST23- KL1.

3.8. Convergence of AMR and virulence

To investigate the relationship between AMR and virulence in *K. pneumoniae*, ESBL (*bla*_{CTX-M-15}, *bla*_{TEM-1B}) and carbapenemase (*bla*_{OXA-232}) harbouring isolates were mapped with major virulence genes (*rmpA*, *magA*, *ybtS*, *allS*, *iutA*). 98% of the ESBL and carbapenemase isolates harboured *ybtS* and *iutA*, 62% of isolates with *rmpA* and *magA*. Hence *K. pneumoniae* seeks prominent attention in clinical settings (Supplementary Table S5).

3.9. Frequency of virulence and AMR genes in *cKp* and *hVkp* isolates

From the statistical analysis, the occurrences of virulence genes are significantly different between hypervirulent and classical strains of *K. pneumoniae*, which clearly helps to categorise the virulent strains, whereas the distributions of AMR genes are overlapping with both the categories (Fig. 6a and 6b).

4. Discussion

This is the first comparative study of *K. pneumoniae* genomes from India, which includes WGS of 144 clinical and 9 environmental strains collected during 2010–2020. Here, we examined antibiotic resistance, diversity of STs, serotypes, virulence genes, plasmids and evolutionary relatedness using PATRIC. A similar kind of epidemiological surveillance of bacterial pathogens is available in BacWGSTdb 2.0 [33].

Emerging drug-resistant and virulent phenotypes of *K. pneumoniae* is a challenging concern worldwide. The prevalence of ESBL-producing microorganism is increasing [34,35]. Nursing homes and intensive care units are the reservoirs of ESBL-harbouring microorganisms, mainly causing sepsis, urinary tract infections, peritonitis, intraabdominal abscesses and upper respiratory infections [36,37].



The present analysis revealed the abundance of bloodstream infections among the Indian population in support of previous report [24].

a Category Hypervirulent strains Classical strains

Wilcoxon, $p = 0.81$

No. of AMR genes

Hypervirulent strains Classical strains

Category

b Category Hypervirulent strains Classical strains

Wilcoxon, $p = 7.3e-11$

No. of virulence gene

Hypervirulent strains Classical strains

Category

(sulfhydryl variable), CTX-M (cefotaximase) and TEM (Temoniera) [40]. Among these, CTX-M-type ESBLs are now found to be the most common type [41]. The *bla*_{CTX-M-15} was widely distributed in clinical isolates since 2013 and increased over time from 5.45% to 30.43% during 2017. As a result, *bla*_{CTX-M-15} renders high resistance to large β -lactam agents. The OXA type ESBLs are found in 125 strains with the highest occurrence of *bla*_{OXA232} (n=66). These *bla*_{OXA232} producers belong to ST231 (n=31), ST14 (n=8) and ST2096 (n=8). The coexistence of *bla*_{NDM} and *bla*_{OXA} was reported to have resistance to all β -lactam antibiotics [42]. In the present analysis we observed the associated *bla*_{NDM} and *bla*_{OXA} in 22.22% of clinical strains [24].

Capsular polysaccharide is the major virulence factor, which protects the bacteria from intracellular killing, phagocytosis, serum complement proteins and oxidative stress conditions [43,44]. Looking across the serotype data set of 153 genomes, we found 29 serotypes among the *K. pneumoniae* isolates, of which K51 (28.1%) and K64 (21.56%) were majorly found. Most of the K51 isolates belong to ST231 (93.02 %) and K64, which are distributed diversely among major clonal types of ST14, ST147, ST395, ST2096 and ST231. Capsular types KL1, KL2, K5 K20 K57 are considered markers for hvKp strains [13,45]. However, they are less prevalent among our study population. In line with previous report KL1 and KL2 isolates

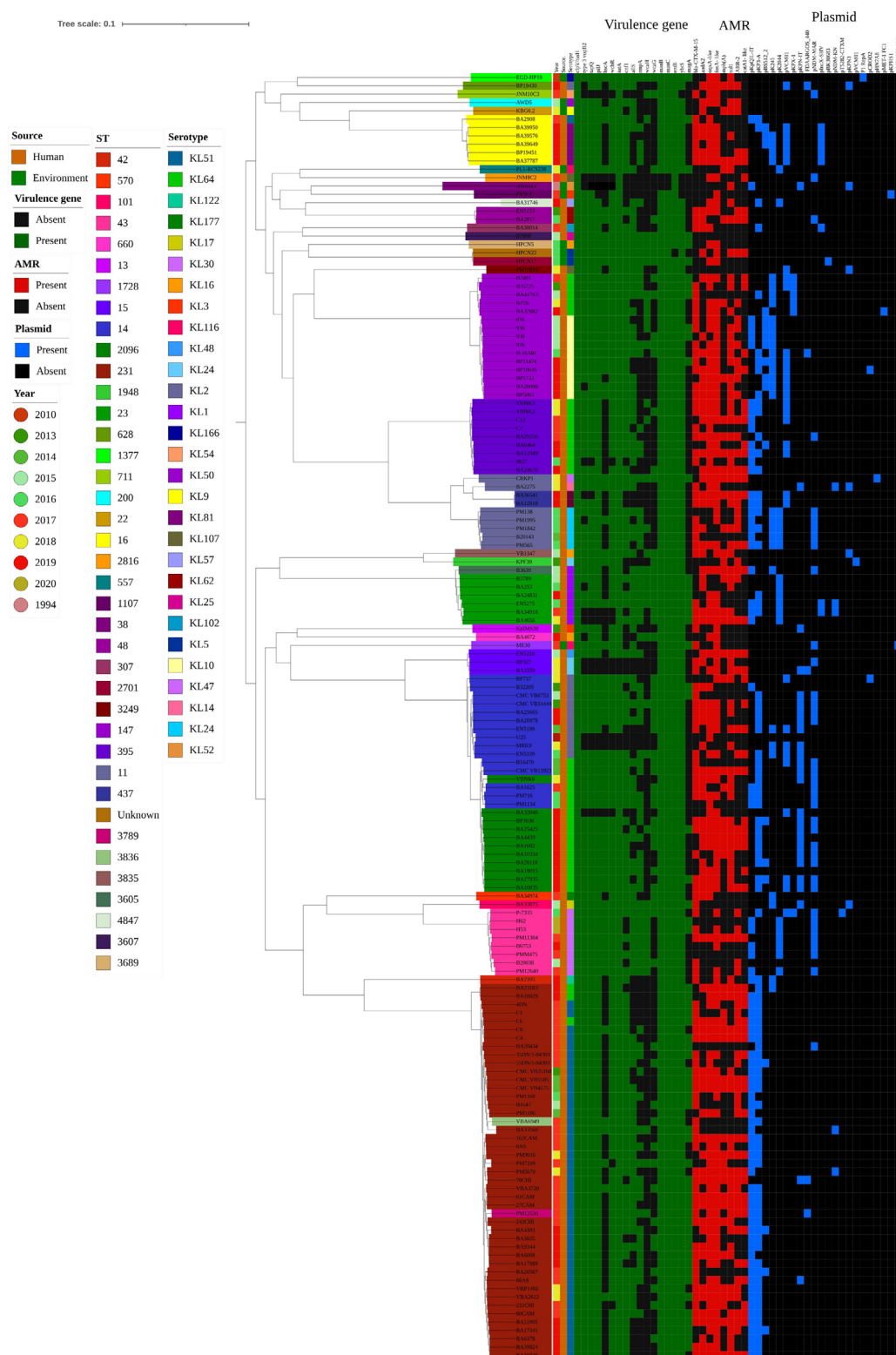


Fig. 7. A complete picture of SNP-based phylogeny: The tree consists of diverse STs and a few major STs that were more frequent in the clinical strains and uncommon STs in environmental strains. The colour of each branch denotes particular sequence typing. The isolation year, source and serotype are indicated by different colours. The presence of AMR genes, virulence genes and replicon types are indicated.

belong to ST23 ($n=6$) and ST14 ($n=10$), respectively. The environmental strains have identified with non-major serotypes such as KL16 ($n=3$), KL3 ($n=2$), KL116 ($n=2$), KL9 ($n=1$) and KL166 ($n=1$). We then correlated the presence of *rpmA* and *magA* with capsular serotypes, where K1 strains ($n=7$) were found to have both *rpmA*

and *magA* ($n=6$), and five K2 strains were not detected with *rpmA* and *magA*. This observation suggests that K1/K2 isolates may partially correspond to hypervirulence. Also, more than 50% of KL51 strains were found to have both *rpmA* and *magA*, recommending the hypervirulence characteristics of KL51.

Other chromosomally encoded pathogenicity factors that enable the bacteria to establish infection and progression of disease by evolving immune evasion strategies were included. These include type 1 fimbriae, type 3 fimbriae, type VI secretion system, type III secretion system translocator protein, type IV pilus secretion, type IV pili, ferric aerobactin receptor, cytotoxic necrotizing factor 1, and allantoin utilization [46]. Enterobactin (EntB), yersiniabactin (YbtS) and aerobactin (iutA) are the prevalent siderophores among *K. pneumoniae* [47]. They promote growth in the upper respiratory tract, lungs and serum. *K. pneumoniae* producing both ybtS and entB can cause pneumonia. In our study, both ybtS and entB genes are prevalent among both clinical and environmental strains, suggesting their role in acquisition of nutrients from the environment. Also, allantoin utilization (allS) is considered an important virulence determinant of *K. pneumoniae* and characterised as hvKp that cause pyogenic liver abscesses [48,49]. In our study population, allS is present in 95 clinical strains and 7 environmental strains.

Similarly, the other virulence genes (type1 fimbriae, T3SS-translocator protein, fimbriae, PilQ, T4SS-pili) were present among all the study strains. The number of associated virulence genes (*rmpA*, *magA*, *entB*, *ybtS*, *iutA*, *allS*) appeared to be higher in ST231-KL51 isolates and ST23-KL1 isolates. Environmental strains (n=3) were found to have all the associated virulence genes.

Discrimination of environmental and clinical strains is important in diagnosis and treatment. We categorise these strains as true environment (root nodule, endophyte), and others may originate from humans or animals (domestic sewage, industrial waste water, river surface water, STP sludge, rhizosphere soil, agriculture field).

The genetic relationship of the isolates was analysed based on SNP and MLST. Environmental isolates were outgrouped (Fig. 7). The MLST typing of environmental strains showed a diversity of sequence type, ST22, ST200, ST557, ST1107, ST1728, ST3689, ST1377, ST2701. These STs, except ST1377, were also reported to be associated in clinical strains [29].

Among these environmental strains, EDG-HP19, the industrial waste water isolate was found to have a high number of resistant genes (AMR genes = 9) when compared with other strains. However, we have noticed less AMR genes (<10) in clinical strains as well (n=20). *oqx-A* like, *oqx-B*-like and *fosA*-like are commonly distributed among the nine strains. The environmental strains have identified with non-major type serotypes such as KL16, KL3, KL116, KL9, KL166 and KL50. The root nodule and domestic sewage isolates have KL5 serotype, which is also considered a marker for hvKp. AWD5 and HPCN22 (n=2) were not detected with *rmpA* and *magA* gene. However, 49 clinical strains were also not detected with these two major virulent genes (Fig. 5). Rhizosphere soil AWD5 *K. pneumoniae* has been shown to be avirulent in a lung infection mice model [50]. As in previous reports, AMR and virulence characteristics are overlapped in both clinical and environment.

Although the genomes of environmental strains collected in this study are not sufficiently conclusive to discriminate the *K. pneumoniae* from clinical settings and soil environments, some interesting observations were made. The phylogenetic analysis clearly separates the environmental from clinical strains, and we have noticed that the environmental strains are characterised by uncommon STs and serotypes except KL5. The plasmids pKpQIL-IT and colKP3 dominant in clinical strains are not detected in environmental strains.

In summary, our analysis with 153 genomes of *K. pneumoniae* revealed the abundance of bloodstream infections among the Indian population. Most commonly, β -lactamase genes *bla*_{CTX-M-15}, *bla*_{TEM-1B} and *bla*_{OXA-232} were found among the study strains. *bla*_{NDM-5} exhibited co-positivity with *bla*_{CTX-M-15} and *bla*_{TEM-1B}. Also, an increasing trend in strains carrying *bla*_{CTX-M-15}, *bla*_{TEM-1B}, and *bla*_{NDM-5} were observed from 2019. The OXA type ESBLs are

found in 124 strains with the highest occurrence of *bla*_{OXA-232} majorly belonging to ST231. Antibiotic efflux is the major drug resistance mechanism among the study strains. IncF conjugative plasmid is abundant followed by pKpQIL-IT and colKP3. These dominant plasmids are not found in environmental strains. The clinical isolates that have both *rmpA* and *magA* were identified as blood isolates majorly with K51 serotype. Diverse ST types were noticed, among which ST231, ST14 and ST147 are prevalent. The number of associated virulence genes (*rmpA*, *magA*, *entB*, *ybtS*, *iutA*, *iutA*, *allS*) was found to be higher in ST231-KL51 and ST23-KL1 isolates, whereas the environmental strains are characterised by uncommon STs and serotypes.

5. Conclusion

The present comparative genomic investigations confirm the convergence of resistance and virulence in *K. pneumoniae* as a major concern to be addressed immediately. In addition, we strongly suggest that detection of *K. pneumoniae* strains with genotype of ST231-KL51 and ST23-KL1 needs more attention for immediate treatment and preventing its dissemination.

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Competing interests

There is no conflict of interest.

Ethical approval

The study does not involve any ethical subjects.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jgar.2022.05.006](https://doi.org/10.1016/j.jgar.2022.05.006).

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